

# Certificate of Analysis

## PRODUCT DETAILS

Name

Date

Cell Source and Passage

Autologous Activated Natural Killer cells

Storage Temperature

4 – 8°C

Viable Cell Count

| TEST                           | SPECIFICATION                                        | TEST RESULTS                   | REMARKS                                                                               |
|--------------------------------|------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------|
| Infectious Diseases Screening  | HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR             | Not Detected /<br>Non-Reactive |    |
| Morphology of Cells            | Round to elongated polymorphic shape                 | Mostly round cells             |    |
| Cell Viability                 | ≥90% ± 0.1 of requested cells                        | Within range                   |  |
| Sterility                      | No cloudiness of the test media                      | Negative                       |  |
| Bacterial Endotoxin Test (LAL) | Endotoxin sensitivity 0.125 EU/mL                    | Negative                       |  |
| Mycoplasma                     | Mean of absorbance <0.05                             | Negative                       |  |
| Immunophenotyping              | Positive Marker – CD16/CD56<br>Negative Marker – CD3 | Detected                       |  |

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Authorised Signature:

Quality Control Department  
Date:

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## Supporting Information

### MORPHOLOGY OF CELLS <sup>1</sup>

Figure 1: Natural Killer cells

### STERILITY TEST <sup>6, 8</sup>

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.

Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

### BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST <sup>2,6</sup>

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

### MYCOPLASMA TEST <sup>3,6</sup>

Negative findings for mycoplasma species *M. hyorhins*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

### IMMUNOPHENOTYPING <sup>4,7</sup>

| Marker | Positive (CD16/56+) | Negative (CD3+) |
|--------|---------------------|-----------------|
| Result | +                   | +               |

‘+’ indicates the presence of the marker in the cell.



### REFERENCES

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2. Bacterial Endotoxins. European Pharmacopoeia 5.0. <https://gmpus.com/validation/Method/LAL/EUPHARMACOPDEIA.pdf>  
3. US Pharmacopeia <8> Mycoplasma Tests  
4. William G. Morice. 2007. The Immunophenotypic Attributes of NK Cells and NK-Cell Lineage Lymphoproliferative Disorders.  
5. USP Method: Sterilization of Medical Devices, Microbiological Methods  
6. ICH Harmonised Tripartite Guideline, Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q6B.  
7. Bryceson YT, Taurat C, Nunes JM, Wood SM, Björkström NK, Long ED, Ljunggren HG. Functional analysis of human NK cells by flow cytometry. Methods Mol Biol. 2010;612:335-52. doi: 10.1007/978-1-60761-962-6\_23. PMID: 20033652. PMID: PMC4869920.  
8. US Pharmacopeia <71> Sterility Test